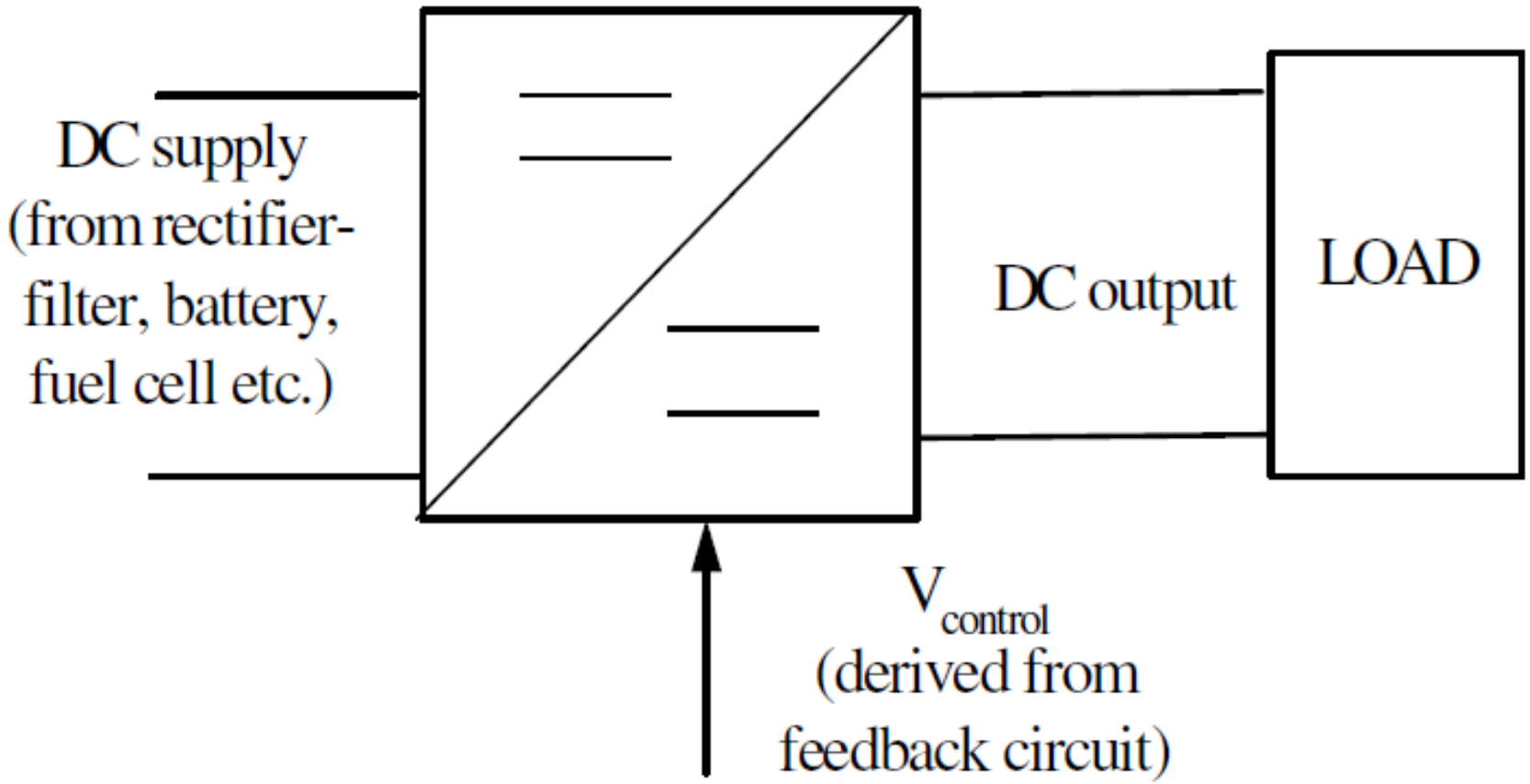
DC-Dc Converter

DC Chopper

General block diagram

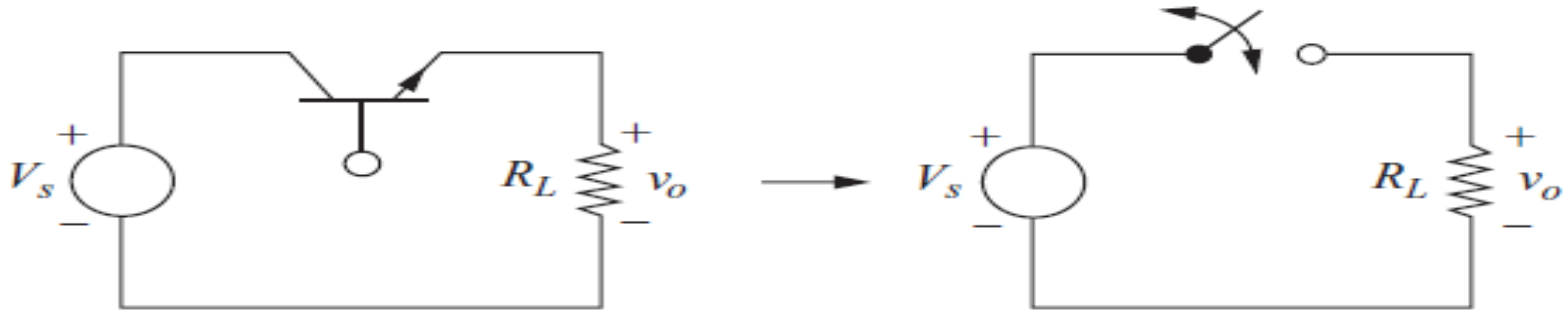


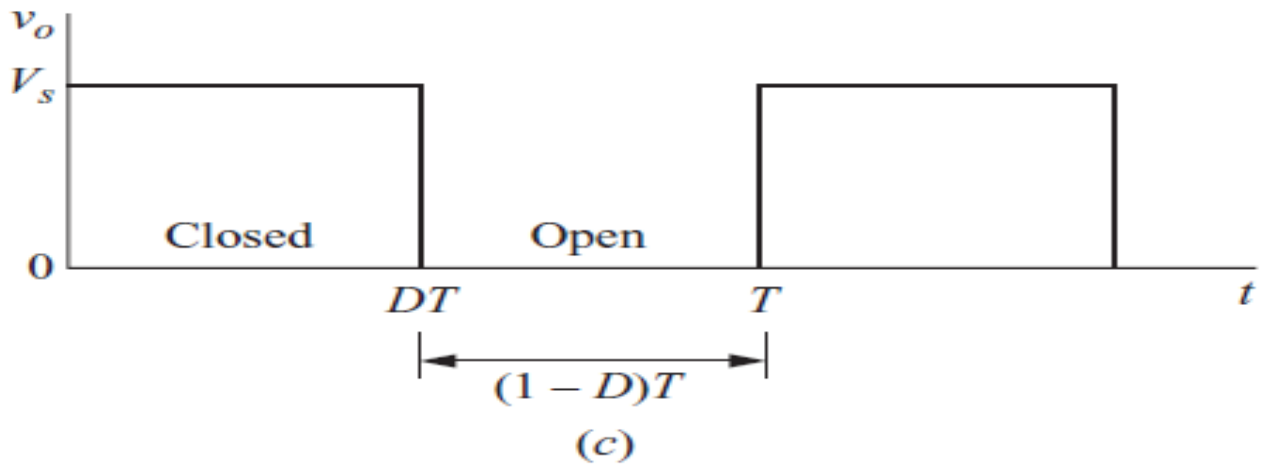
Introduction

- Chopper is a static device.
- A variable dc voltage is obtained from a constant dc voltage source.
- Also known as dc-to-dc converter.
- Widely used for motor control.
- Also used in regenerative braking.
- Thyristor converter offers greater efficiency, faster response, lower maintenance, smaller size and smooth control.

A Basic Switching converter

- In a switching converter circuit, the transistor operates as an electronic switch by being completely on or completely off (saturation or cutoff for a BJT or the triode and cutoff regions of a MOSFET). This circuit is also known as a DC chopper.
- Assuming the switch is ideal in Fig. 6-2, the output is the same as the input when the switch is closed, and the output is zero when the switch is open. Periodic opening and closing of the switch results in the pulse output shown in Fig. 6-2c.


+
V_s
−
R_L
+
v_o
−
(a)
+
V_s
−
R_L
+
v_o
−
(b)


v_o
V_s
0
Closed
Open
DT
T
t
$(1 - D)T$
(c)

The average or DC component of the output voltage is:

$$V_o = \frac{1}{T}\int_0^T v_o(t)\,dt = \frac{1}{T}\int_0^{DT} V_s\,dt = V_s D \qquad (6\text{-}1)$$

The dc component of the output voltage is controlled by adjusting the duty ratio *D*, which is the fraction of the switching period that the switch is closed

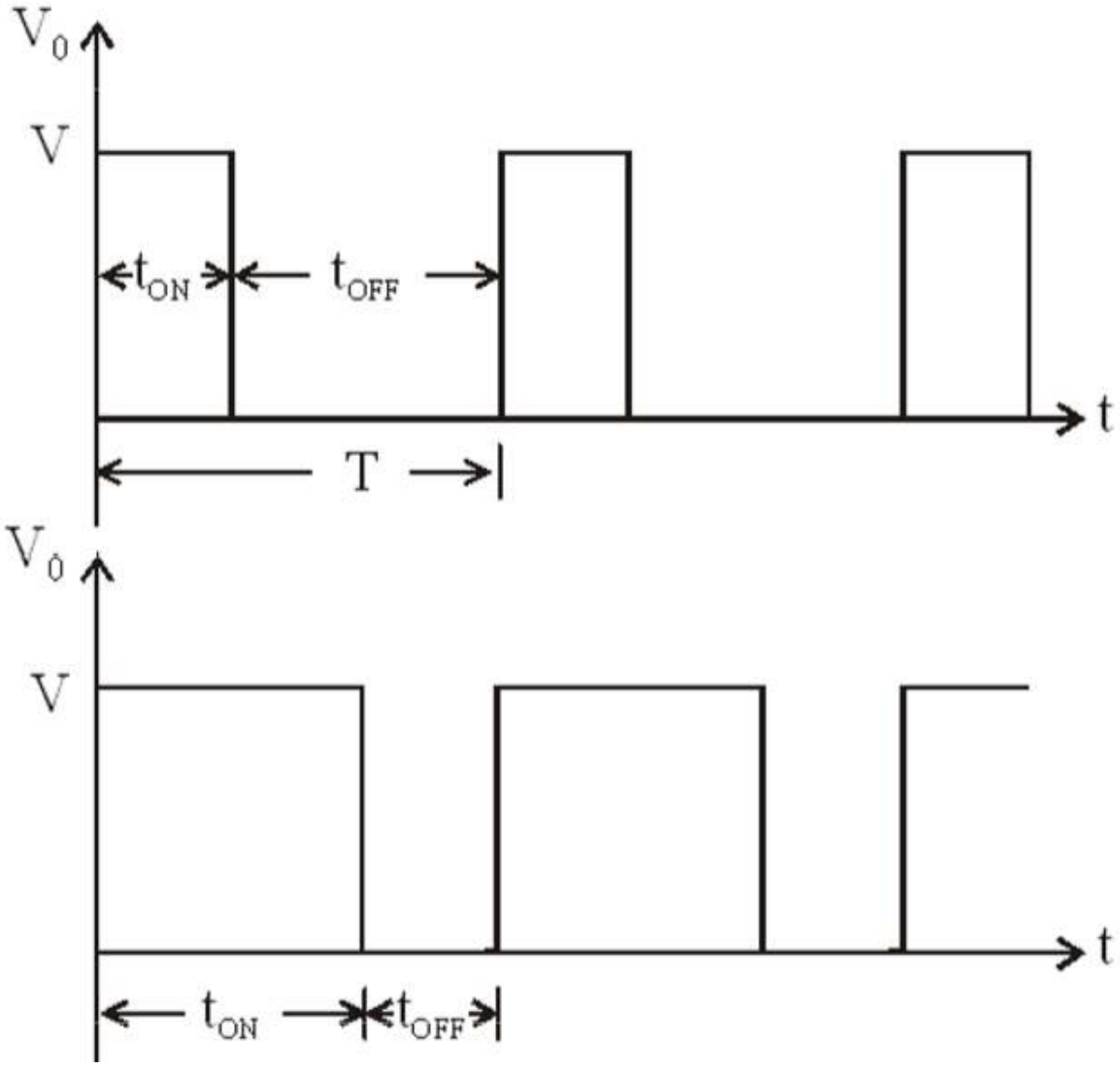
$$D \equiv \frac{t_{on}}{t_{on} + t_{off}} = \frac{t_{on}}{T} = t_{on} f \qquad (6\text{-}2)$$

Where *f* is the switching frequency. The DC component of the output voltage will be less than or equal to the input voltage for this circuit.

Pulse Width Modulation

- t_{ON} is varied keeping chopping frequency '*f*' & chopping period 'T' constant.
- Output voltage is varied by varying the ON time t_{ON}

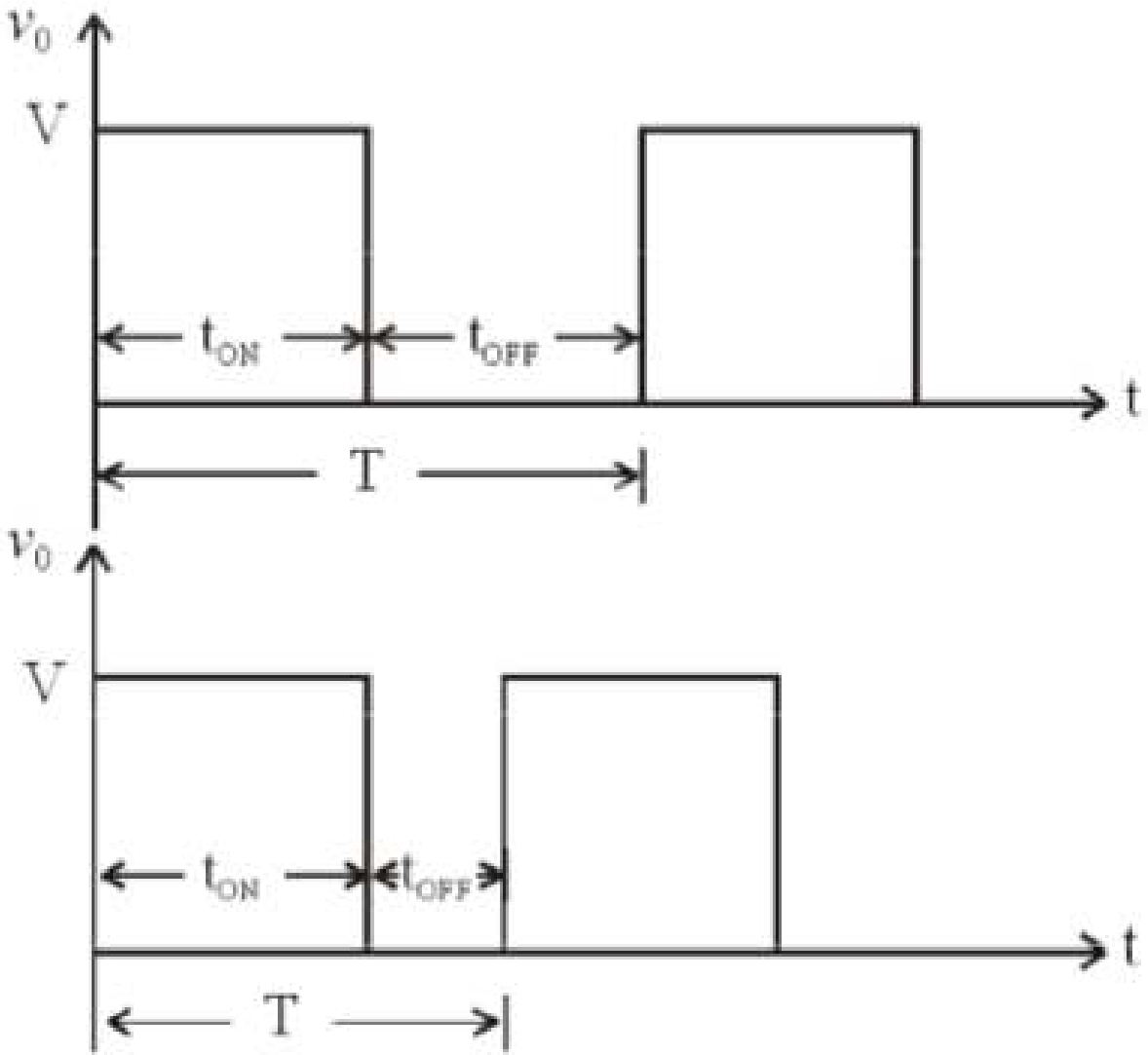
Pulse Width Modulation Control



Variable Frequency Control

- Chopping frequency *'f'* is varied keeping either t_{ON} or t_{OFF} constant.
- To obtain full output voltage range, frequency has to be varied over a wide range.
- This method produces harmonics in the output and for large t_{OFF} load current may become discontinuous

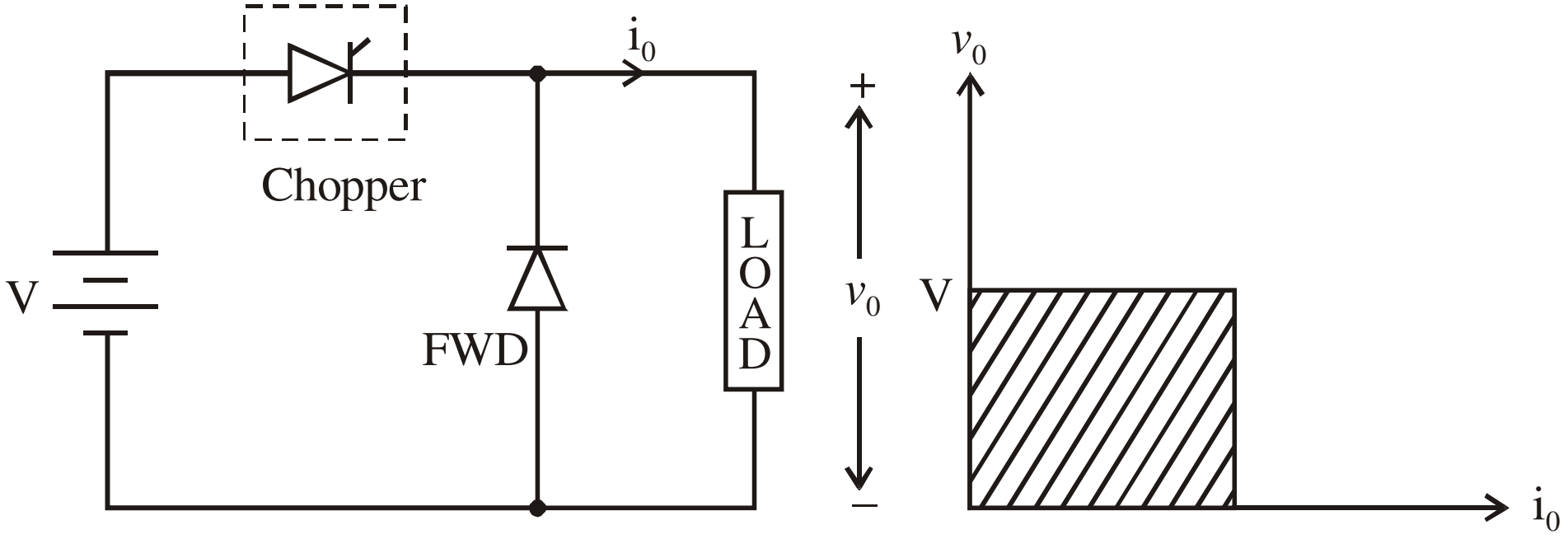
Frequency Modulation



DC Choppers Classification

- Choppers are classified as
  - Class A Chopper
  - Class B Chopper
  - Class C Chopper
  - Class D Chopper
  - Class E Chopper

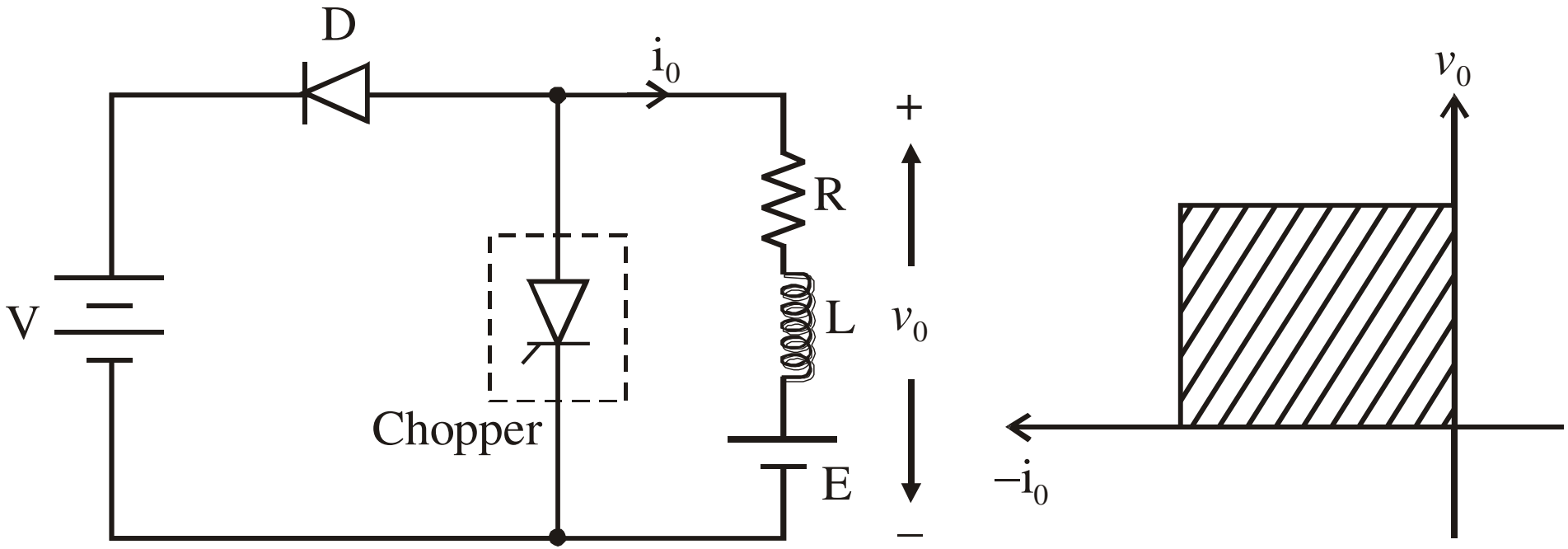
Class A Chopper/First-quadrant/Buck (step-down) converter



- *Class A Chopper* is a step-down chopper in which power always flows form source to load.

- It is used to control the speed of a DC motor.

Class B Chopper/second quadrant /Boost (step-up) converter



- When chopper is ON, E drives a current through L and R in a direction opposite to that shown in figure.

- During the ON period of the chopper, the inductance L stores energy.

- When Chopper is OFF, diode D conducts, and part of the energy stored in inductor L is returned to the supply.

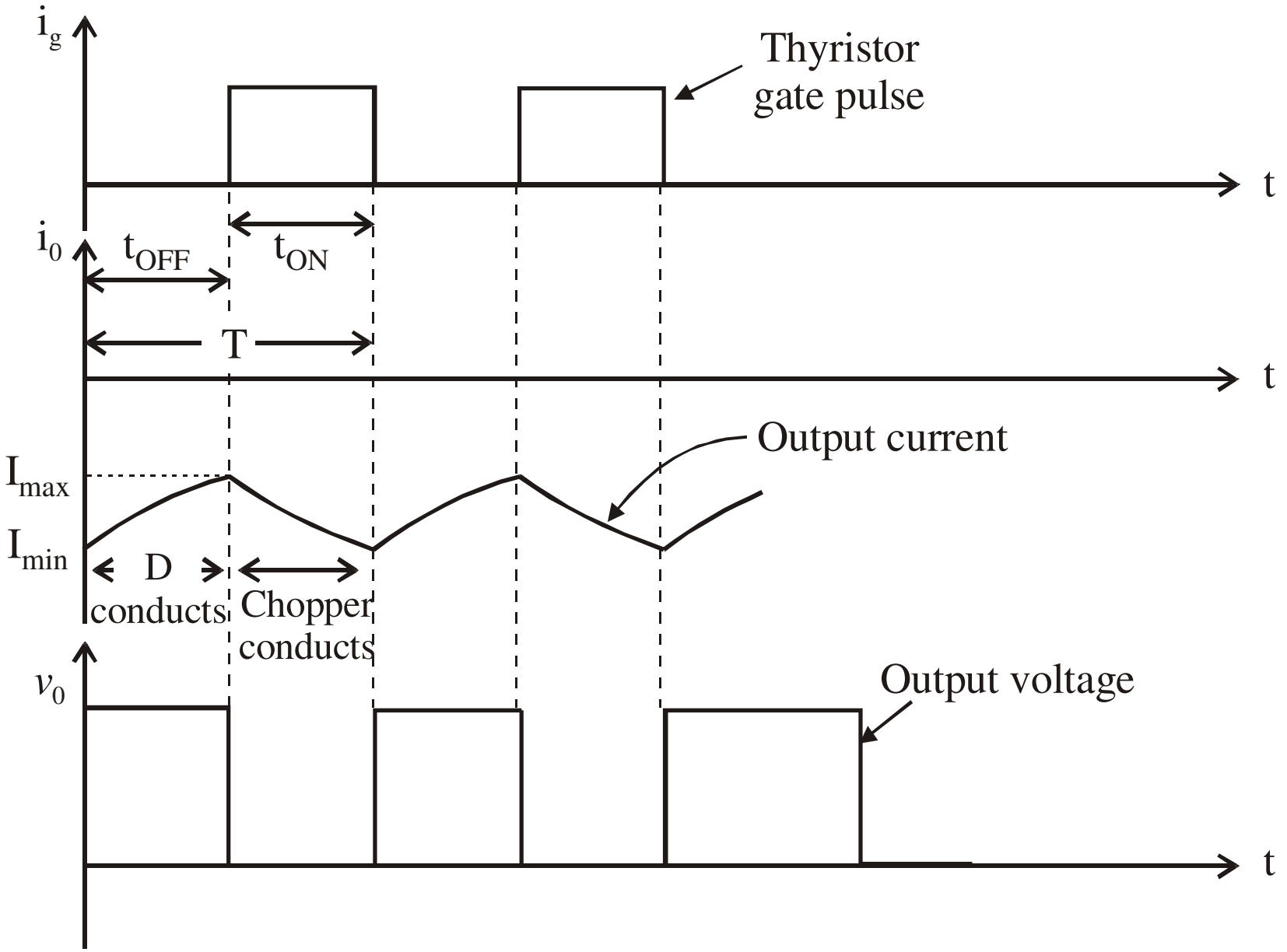
- Average output voltage is positive.
- Average output current is negative.

- Therefore C*lass B Chopper* operates in second quadrant.

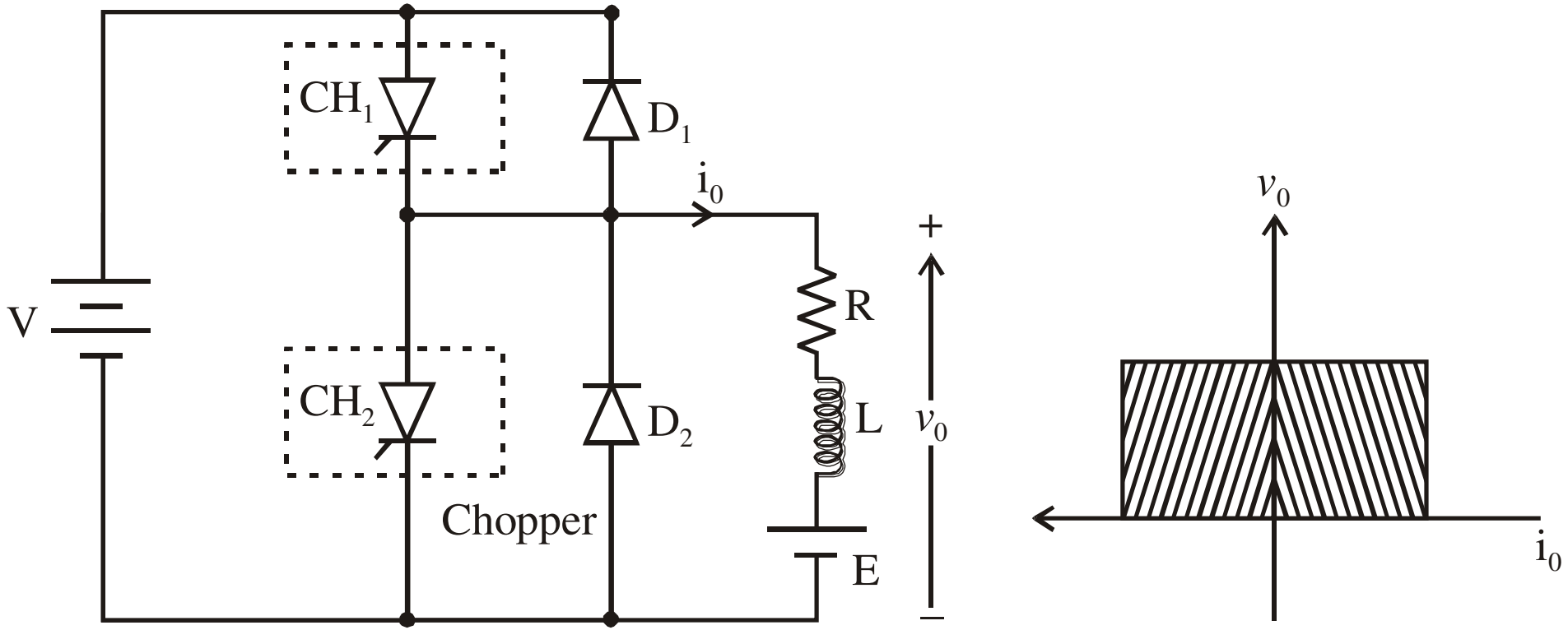
- In this chopper, power flows from load to source.

- *Class B Chopper* is used for regenerative braking of DC motor.

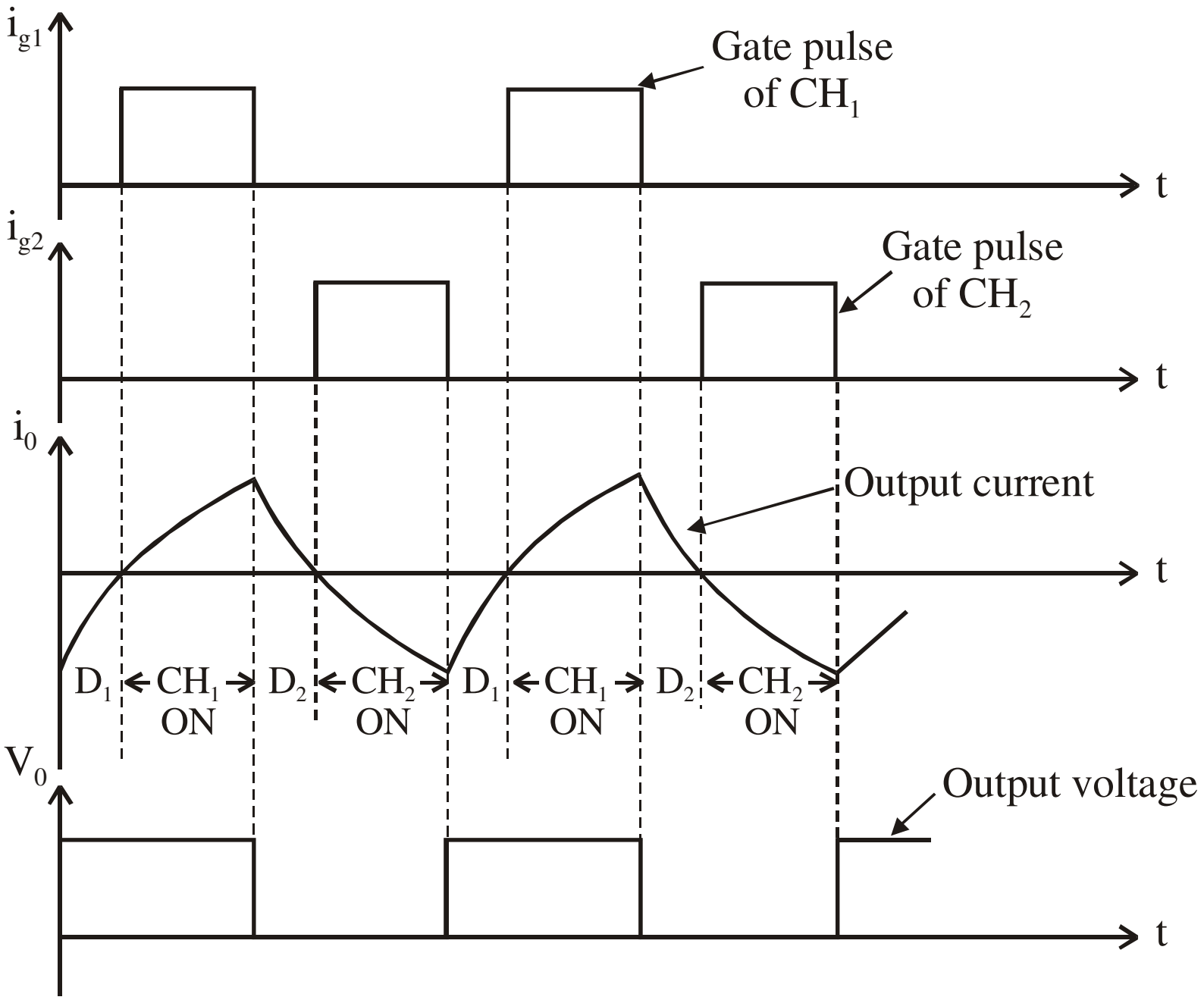
- *Class B Chopper* is a step-up chopper.


i_g
Thyristor
gate pulse
t
i_0
t_{OFF}
t_{ON}
T
t
Output current
I_{max}
I_{min}
D
conducts
Chopper
conducts
v_0
Output voltage
t

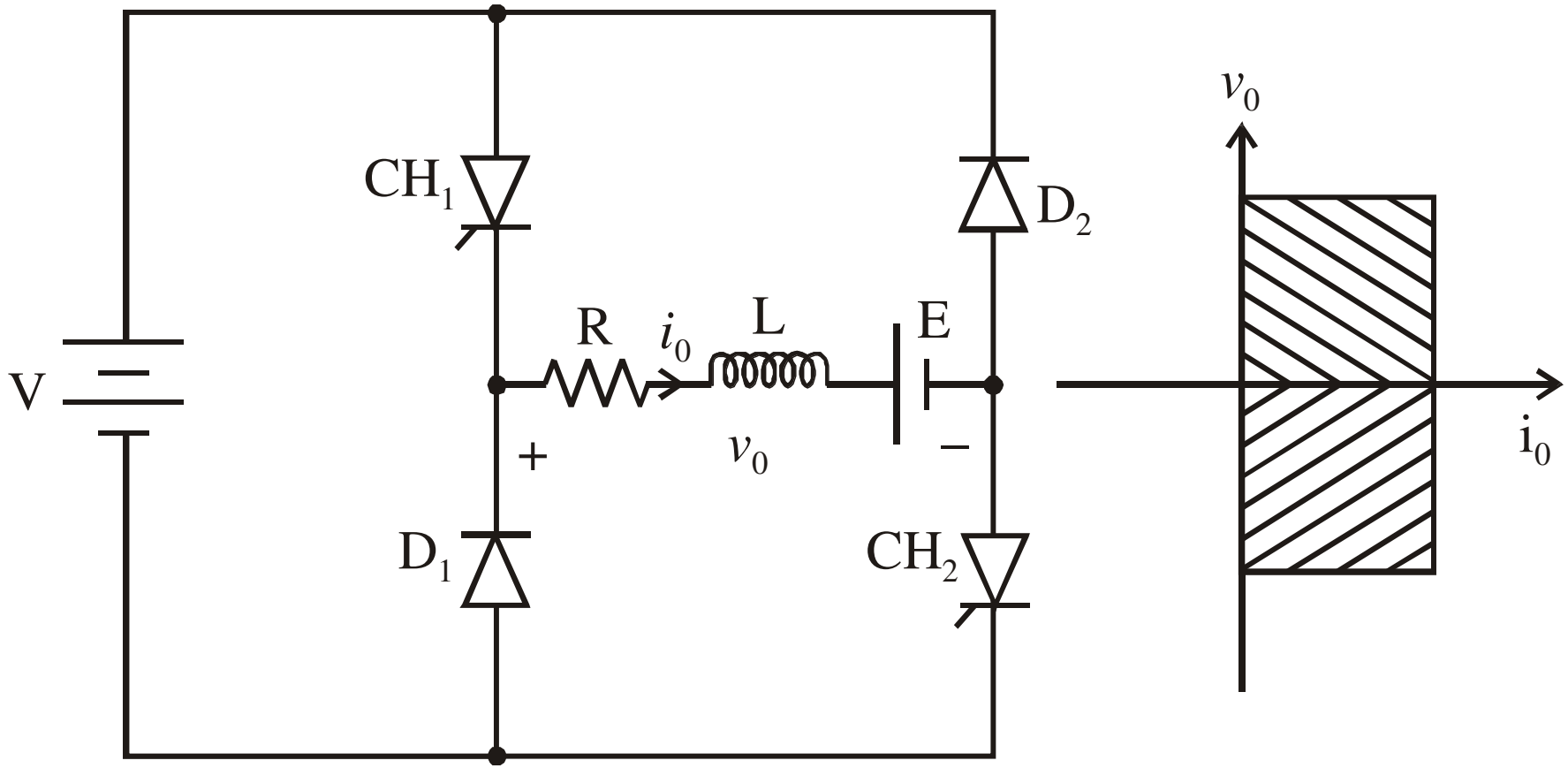
Class C Chopper/Two-quadrant type-A/**Step Up Step Down /**Buck-Boost converter



- Current continues to flow in positive direction.
- When CH_2 is triggered, the voltage E forces current to flow in opposite direction through L and CH_2.
- The output voltage is zero.
- On turning OFF CH_2, the energy stored in the inductance drives current through diode D_1 and the supply
- Output voltage is V, the input current becomes negative and power flows from load to source.


i_{g1}
Gate pulse of CH_1
t
i_{g2}
Gate pulse of CH_2
t
i_0
Output current
t
D_1 CH_1 ON
D_2 CH_2 ON
D_1 CH_1 ON
D_2 CH_2 ON
V_0
Output voltage
t

Class D Chopper/**Two-Quadrant Type B**



- Class D is a two quadrant chopper.

- When both CH_1 and CH_2 are triggered simultaneously, the output voltage $v_O = V$ and output current flows through the load.

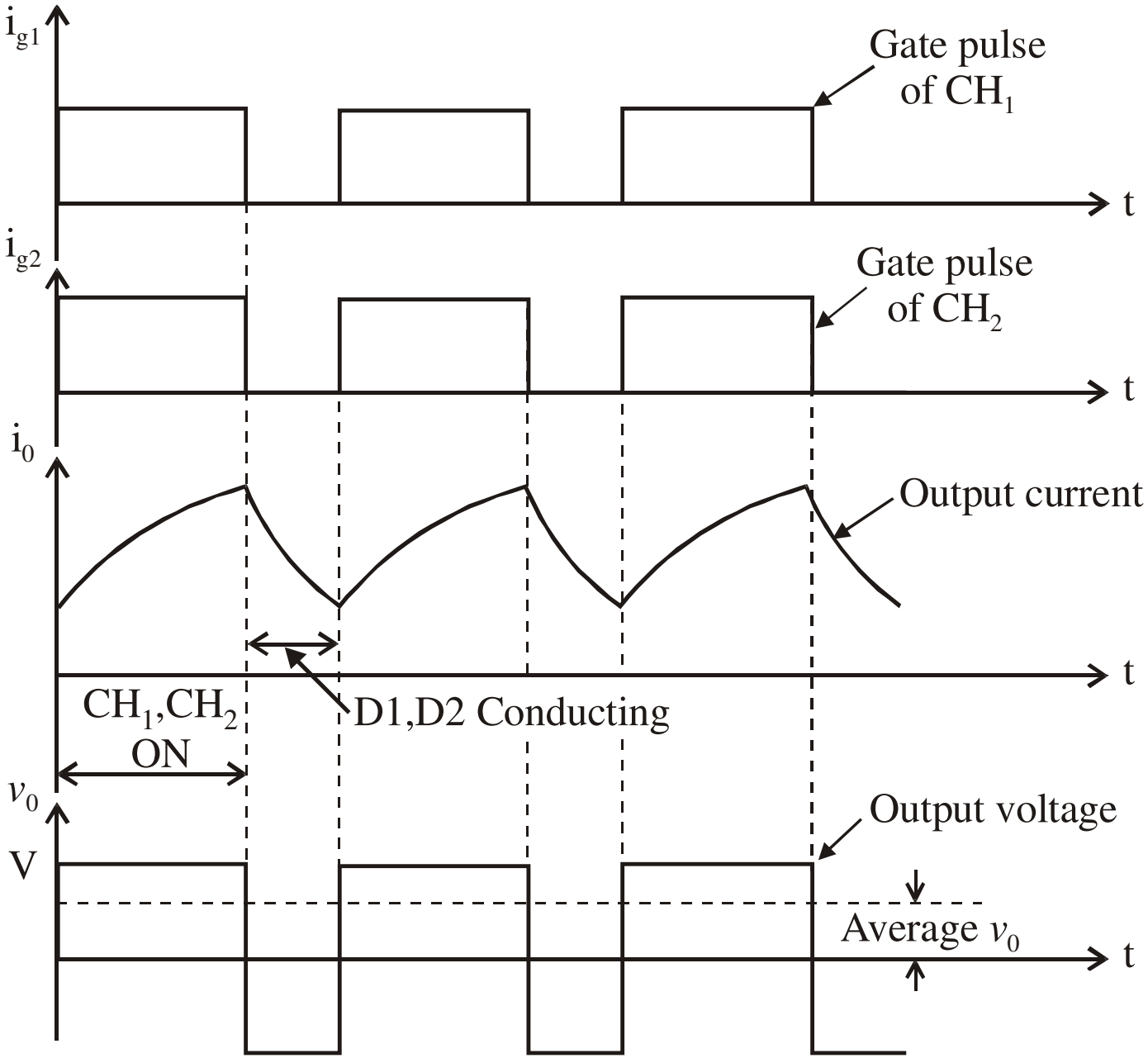
- When CH_1 and CH_2 are turned OFF, the load current continues to flow in the same direction through load, D_1 and D_2, due to the energy stored in the inductor L.

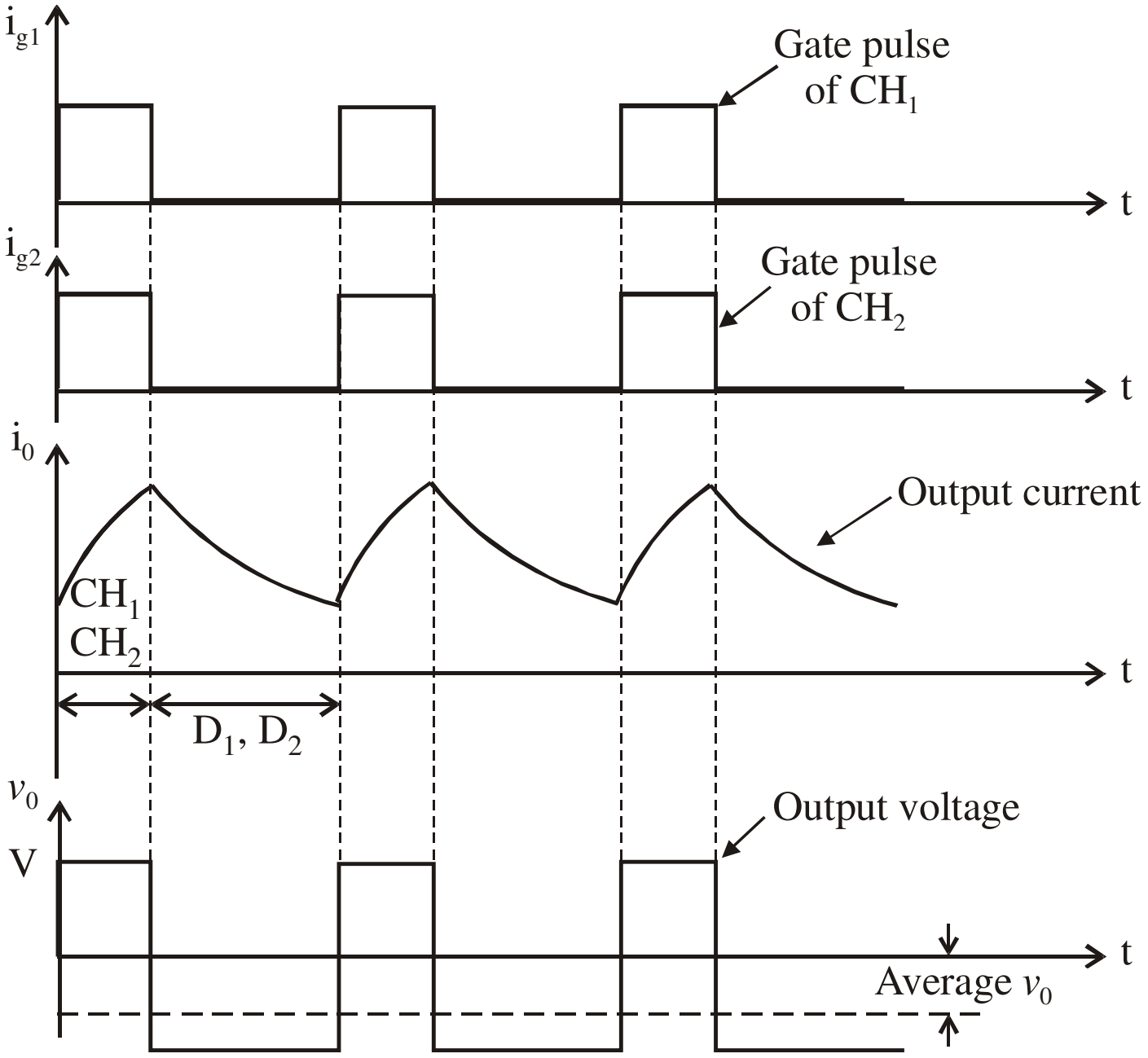
- Output voltage $v_O = -V$.

- Average load voltage is positive if chopper ON time is more than the OFF time.

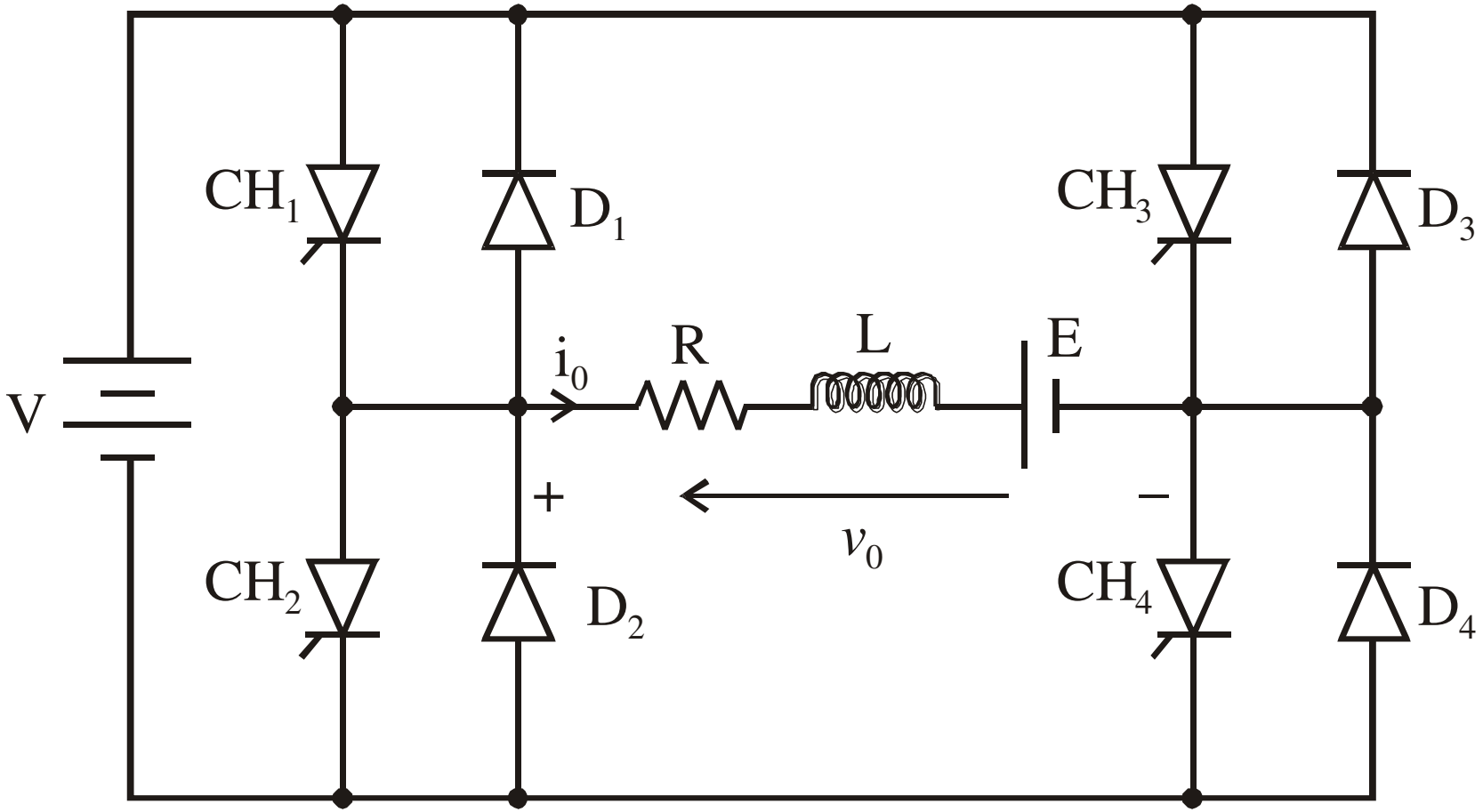
- Average output voltage becomes negative if $t_{ON} < t_{OFF}$.

- Hence the direction of load current is always positive but load voltage can be positive or negative.

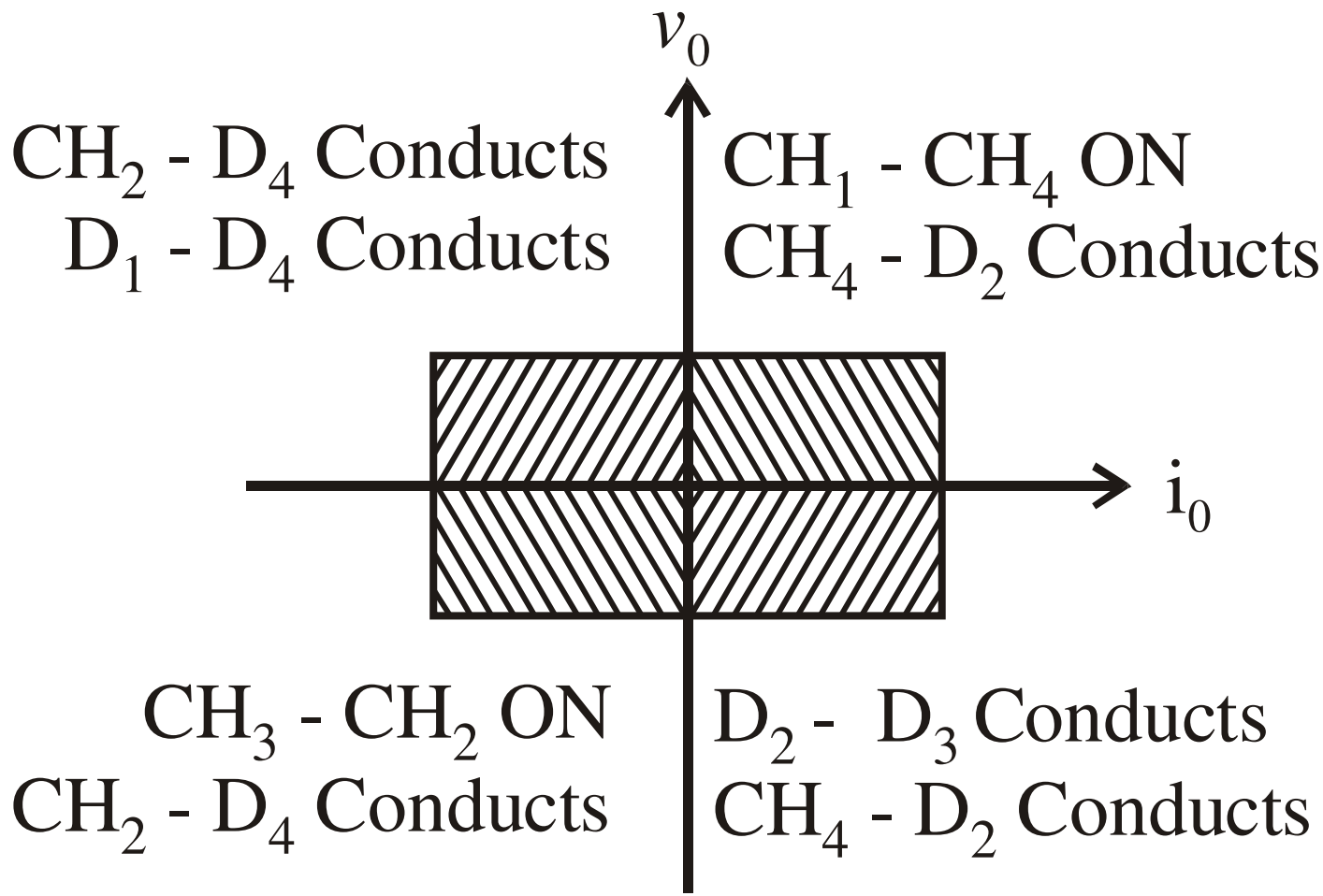

i_{g1}
Gate pulse of CH_1
t
i_{g2}
Gate pulse of CH_2
t
i_0
Output current
t
CH_1,CH_2 ON
D1,D2 Conducting
v_0
V
Output voltage
Average v_0
t


i_{g1}
Gate pulse of CH_1
t
i_{g2}
Gate pulse of CH_2
t
i_0
Output current
CH_1
CH_2
t
D_1, D_2
v_0
Output voltage
V
t
Average v_0

Class E Chopper/**Fourth-Quadrant**



Four Quadrant Operation



- Class E is a four quadrant chopper.

- When CH_1 and CH_4 are triggered, output current i_O flows in positive direction through CH_1 and CH_4, and with output voltage $v_O = V$.

- This gives the first quadrant operation.

- When both CH_1 and CH_4 are OFF, the energy stored in the inductor L drives i_O through D_2 and D_3 in the same direction, but output voltage $v_O = -V$.

- Therefore the chopper operates in the fourth quadrant.

- When CH_2 and CH_3 are triggered, the load current i_O flows in opposite direction & output voltage $v_O = -V$.

- Since both i_O and v_O are negative, the chopper operates in third quadrant.

- When both CH_2 and CH_3 are OFF, the load current i_O continues to flow in the same direction D_1 and D_4 and the output voltage $v_O = V$.

- Therefore the chopper operates in second quadrant as v_O is positive but i_O is negative.

The BUCK (Step-Down) Converter-ClassA

The average output voltage is given by:

$$\boxed{V_o = V_s D} \qquad (6\text{-}9)$$

The average inductor current must be the same as the average current in the load resistor:

$$I_L = I_R = \frac{V_o}{R} \qquad (6\text{-}10)$$

Average Output Current

$$I_{dc} = \frac{V_{dc}}{R}$$

$$I_{dc} = \frac{V}{R}\left(\frac{t_{ON}}{T}\right) = \frac{V}{R}d$$

RMS value of output voltage

$$V_O = \sqrt{\frac{1}{T}\int_0^{t_{ON}} v_o^2 dt}$$

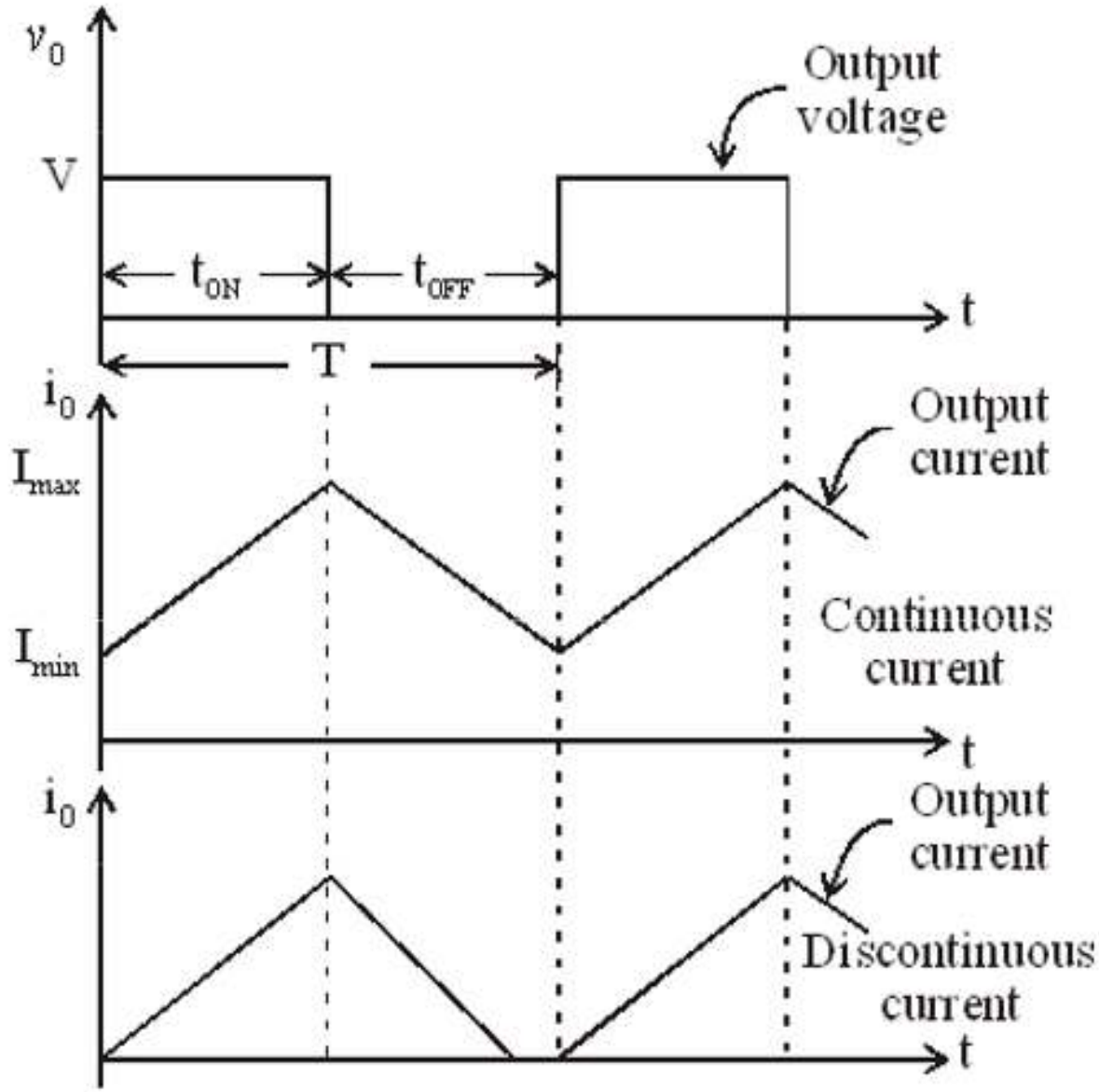
But during t_{ON}, $v_o = V$

Therefore RMS output voltage

$$V_O = \sqrt{\frac{1}{T}\int_0^{t_{ON}} V^2 dt}$$

$$V_O = \sqrt{\frac{V^2}{T} t_{ON}} = \sqrt{\frac{t_{ON}}{T}}.V$$

$$V_O = \sqrt{d}.V$$



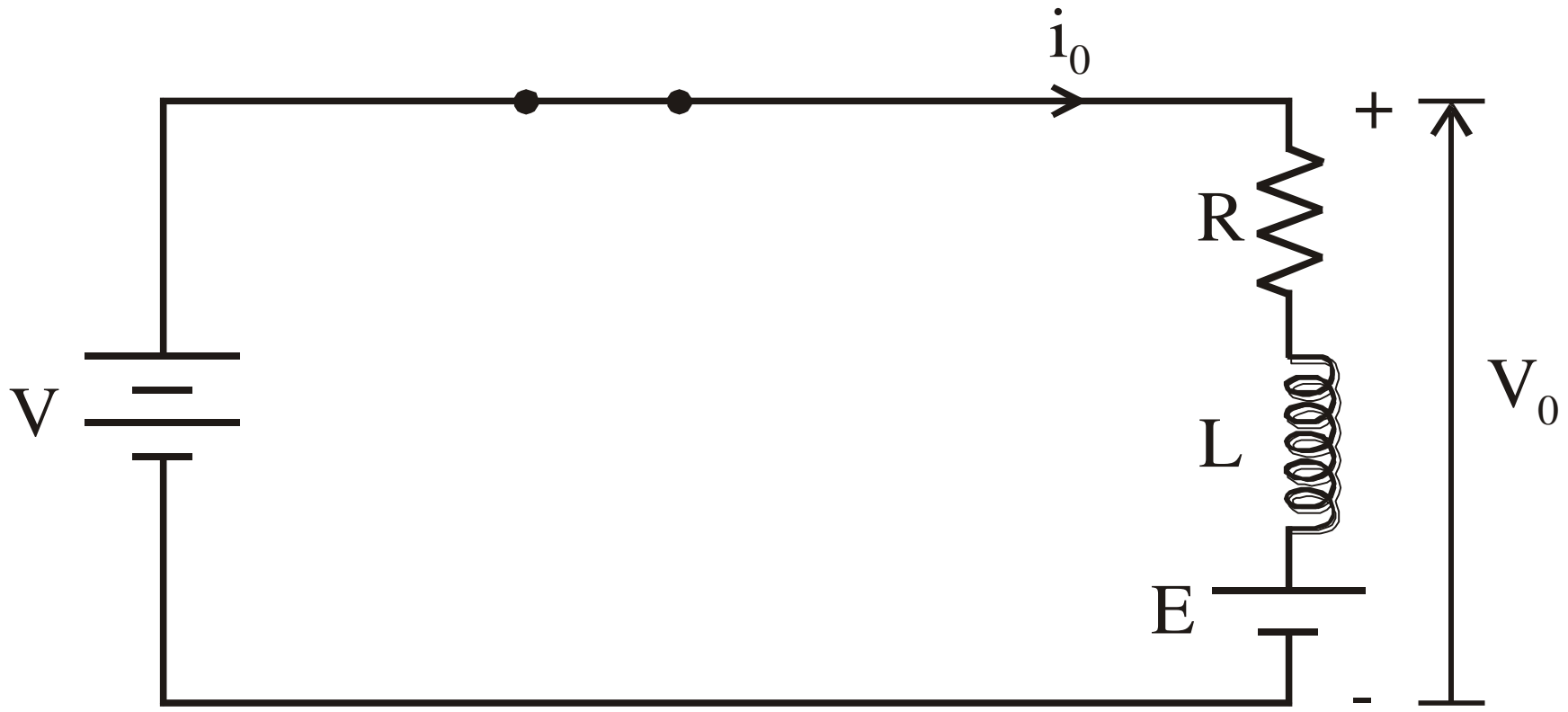
When the current exceeds I_{max} the chopper is turned-off and it is turned-on when current reduces to I_{min} .

Expressions For Load Current

i_O For Continuous Current Operation

When Chopper is ON $(0 \leq t \leq t_{ON})$

When switch is closed:



$$V = i_O R + L\frac{di_O}{dt} + E$$

Taking Laplace Transform

$$\frac{V}{S} = RI_O(S) + L\left[S.I_O(S) - i_O\left(0^-\right)\right] + \frac{E}{S}$$

At $t = 0$, initial current $i_O\left(0^-\right) = I_{\min}$

$$I_O(S) = \frac{V - E}{LS\left(S + \frac{R}{L}\right)} + \frac{I_{\min}}{S + \frac{R}{L}}$$

Taking Inverse Laplace Transform

$$i_O(t) = \frac{V-E}{R}\left[1-e^{-\left(\frac{R}{L}\right)t}\right] + I_{\min} e^{-\left(\frac{R}{L}\right)t}$$

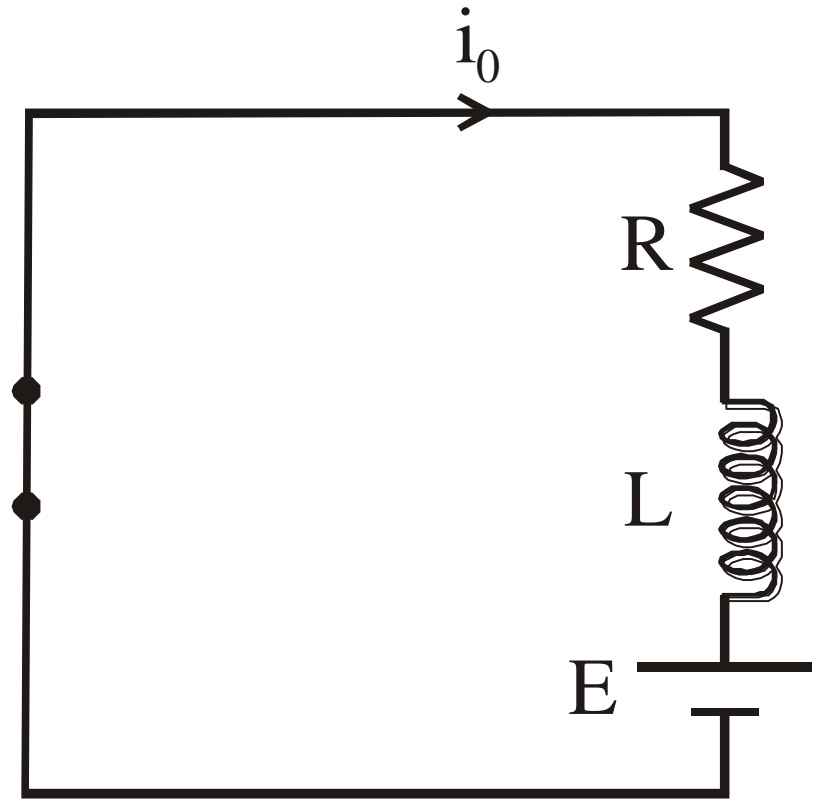
This expression is valid for $0 \le t \le t_{ON}$,

i.e., during the period chopper is ON

At the instant the chopper is turned off,

load current is $i_O(t_{ON}) = I_{\max}$

When Chopper is OFF



When Chopper is OFF $\left(0 \le t \le t_{OFF}\right)$

$$0 = Ri_O + L\frac{di_O}{dt} + E$$

Talking Laplace transform

$$0 = RI_O(S) + L\left[SI_O(S) - i_O\left(0^-\right)\right] + \frac{E}{S}$$

Redefining time origin we have at $t = 0$,

initial current $i_O\left(0^-\right) = I_{\max}$

$$\therefore I_O(S) = \frac{I_{max}}{S + \frac{R}{L}} - \frac{E}{LS\left(S + \frac{R}{L}\right)}$$

Taking Inverse Laplace Transform

$$i_O(t) = I_{max} e^{-\frac{R}{L}t} - \frac{E}{R}\left[1 - e^{-\frac{R}{L}t}\right]$$

The expression is valid for $0 \le t \le t_{OFF}$,

i.e., during the period chopper is OFF

At the instant the chopper is turned ON or at the end of the off period, the load current is

$$i_O(t_{OFF}) = I_{min}$$

To Find I_{max} & I_{min}

From equation

$$i_O(t) = \frac{V - E}{R}\left[1 - e^{-\left(\frac{R}{L}\right)t}\right] + I_{min}e^{-\left(\frac{R}{L}\right)t}$$

At $t = t_{ON} = dT,\ i_O(t) = I_{max}$

$$\therefore \quad I_{max} = \frac{V - E}{R}\left[1 - e^{-\frac{dRT}{L}}\right] + I_{min}e^{-\frac{dRT}{L}}$$

From equation

$$i_O(t) = I_{max} e^{-\frac{R}{L}t} - \frac{E}{R}\left[1 - e^{-\frac{R}{L}t}\right]$$

At $t = t_{OFF} = T - t_{ON}$, $i_O(t) = I_{min}$

$$t = t_{OFF} = (1 - d)T$$

$$\therefore \qquad I_{\min} = I_{\max} e^{-\frac{(1-d)RT}{L}} - \frac{E}{R}\left[1 - e^{-\frac{(1-d)RT}{L}}\right]$$

Substituting for $I_{\min}$ in equation

$$I_{\max} = \frac{V-E}{R}\left[1 - e^{-\frac{dRT}{L}}\right] + I_{\min} e^{-\frac{dRT}{L}}$$

we get,

$$I_{\max} = \frac{V}{R}\left[\frac{1 - e^{-\frac{dRT}{L}}}{1 - e^{-\frac{RT}{L}}}\right] - \frac{E}{R}$$

Substituting for I_{max} in equation

$$I_{min} = I_{max} e^{-\frac{(1-d)RT}{L}} - \frac{E}{R}\left[1 - e^{-\frac{(1-d)RT}{L}}\right]$$

we get,

$$I_{min} = \frac{V}{R}\left[\frac{e^{\frac{dRT}{L}} - 1}{e^{\frac{RT}{L}} - 1}\right] - \frac{E}{R}$$

$(I_{max} - I_{min})$ is known as the steady state ripple.

Therefore peak-to-peak ripple current

$$\Delta I = I_{max} - I_{min}$$

Average output voltage

$$V_{dc} = d.V$$

Average output current

$$I_{dc(approx)} = \frac{I_{max} + I_{min}}{2}$$

Assuming load current varies linearly from I_{min} to I_{max} instantaneous load current is given by

$$i_O = I_{min} + \frac{(\Delta I)t}{dT} \quad for\ 0 \le t \le t_{ON}(dT)$$

$$i_O = I_{min} + \left(\frac{I_{max} - I_{min}}{dT}\right)t$$

RMS value of load current

$$I_{(RMS)} = \sqrt{\frac{1}{dT}\int_0^{dT} i_0^2 dt}$$

$$I_{(RMS)} = \sqrt{\frac{1}{dT}\int_0^{dT}\left[I_{min} + \frac{(I_{max} - I_{min})t}{dT}\right]^2 dt}$$

$$I_{(RMS)} = \sqrt{\frac{1}{dT}\int_0^{dT}\left[I_{min}^2 + \left(\frac{I_{max} - I_{min}}{dT}\right)^2 t^2 + \frac{2I_{min}(I_{max} - I_{min})t}{dT}\right] dt}$$

RMS value of output current

$$I_{O(RMS)} = \left[I_{min}^2 + \frac{(I_{max} - I_{min})^2}{3} + I_{min}(I_{max} - I_{min}) \right]^{\frac{1}{2}}$$

RMS chopper current

$$I_{CH} = \sqrt{\frac{1}{T}\int_0^{dT} i_0^2 dt}$$

$$I_{CH} = \sqrt{\frac{1}{T}\int_0^{dT} \left[I_{min} + \left(\frac{I_{max} - I_{min}}{dT} \right) t \right]^2 dt}$$

$$I_{CH} = \sqrt{d}\left[I_{min}^2 + \frac{(I_{max} - I_{min})^2}{3} + I_{min}(I_{max} - I_{min}) \right]^{\frac{1}{2}}$$

$$I_{CH} = \sqrt{d}\, I_{C(RMS)}$$

Effective input resistance is

$$R_i = \frac{V}{I_S}$$

Where

$$I_S = \text{Average source current}$$

$$I_S = dI_{dc}$$

$$\therefore \quad R_i = \frac{V}{dI_{dc}}$$

The maximum and minimum values of the inductor current are computed as:

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_o}{R} + \frac{1}{2}\left[\frac{V_o}{L}(1-D)T\right] = V_o\left(\frac{1}{R} + \frac{1-D}{2Lf}\right) \quad (6\text{-}11)$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_o}{R} - \frac{1}{2}\left[\frac{V_o}{L}(1-D)T\right] = V_o\left(\frac{1}{R} - \frac{1-D}{2Lf}\right) \quad (6\text{-}12)$$

where $f = 1/T$ is the switching frequency

The Boost Converter-Class B

- The boost converter is shown in Fig. 6-8. This is another switching converter that operates by periodically opening and closing an electronic switch. It is called a boost converter because the output voltage is larger than the input.
- Output Voltage is:

$$V_o = \frac{V_s}{1 - D}$$

Output power is:

$$P_o = \frac{V_o^2}{R} = V_o I_o$$

Maximum and minimum inductor currents are:

$$I_{\max} = I_L + \frac{\Delta i_L}{2} = \frac{V_s}{(1-D)^2 R} + \frac{V_s DT}{2L} \qquad (6\text{-}29)$$

$$I_{\min} = I_L - \frac{\Delta i_L}{2} = \frac{V_s}{(1-D)^2 R} - \frac{V_s DT}{2L} \qquad (6\text{-}30)$$